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No. 26.—VOL. I.

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OUR HEALTH.

BY DR ANDREW WILSON, F.R.S.E.

III. SOME FOOD-DANGERS, AND HOW TO AVOID THEM.

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dairies have been found to be constructed in a thoroughly insanitary manner, and cleanliness—the first condition where milk is concerned—is by no means always observed. The remedy for these errors and negligences in connection with this all-important article of diet, lies in one direction only—namely, a system of rigid and continuous dairy inspection. Such inspection is never complained of by those tradesmen who take a pride in their occupation, and who endeavour, by ordinary attention to business, to secure the purity of the milk they sell. It might be added also, that if other articles of food are duly liable to official examination, and if the articles sold by grocer and butcher are duly supervised and examined, there is no reason why the premises of the dairyman should not be similarly inspected. We do not, as a rule, contract serious illness from impure coffee, or even from a poor quality of butcher-meat; but a dirty dairy and an infected milk-supply may, in a single day, sow the germs of a fever which may prostrate a village or community, and entail all the misery and hardship which serious illness inevitably carries in its train.

The domestic care of milk is a second topic to which the attention of the householder should be directed. It cannot be too clearly borne in mind that milk, of all fluids, is singularly apt to absorb deleterious matters. Sewage-emanations and other gases, paint, metallic matters, &c., are all readily taken up by milk. Hence the absolute necessity for seeing that when milk is received into our homes, it is stored in a safe and sanitary position. Milk should never be stored in metallic vessels in the first place; and it should not be kept in cupboards or other receptacles which are situated in the neighbourhood of sinks, closets, or open drains. Too frequently, such carelessness in the home-treatment of the milk-supply leads to illness, which is all the more serious, because its origin is unsuspected.

With regard to the liability of milk, taken from cows suffering from various diseases, to produce illness in man, many and varied opinions exist. A general rule, and one in the observance of which great safety exists, is, that milk from an animal in any way affected with disease should never be sold to the public. Where uncertainty exists, it is a matter of sheer common-sense to err on the safe side, and to incur no risk whatever. It is only fair to add, that milk from cows suffering from ‘foot-and-mouth’ disease has been consumed in many cases without injury resulting. But opposed to this fact, we find cases in which the use of such milk has been followed by throat-

ailments and other troubles in man. The milk of over-driven cows—‘heated milk,’ as it is called—has been known to produce colic and diarrhœa in children. It is also probable that while some persons in robust health may escape, others are liable to be affected by milk taken from diseased animals. Pigs to which the milk of cows, ill with ‘foot-and-mouth’ disease, has been given, are seized with that disease in a few hours. The safe rule, therefore, appears to be that already mentioned. If a cow is affected with any disorder or disease, the milk of the animal should not be consumed by man. Only by attention to this rule can outbreaks of disease in man be avoided, and the public safety fully secured.

The flesh of animals is liable to acquire under certain conditions diseased properties. Hence, it is necessary that we should be on our guard against such sources of illness. Thus, certain fevers to which pigs, sheep, and cattle are subject render their flesh unfit for human food; and there are certain parasites inhabiting the flesh of fish which may also be productive of disease when the meat in question has been eaten by man.

Good meat in a fresh state should be firm and elastic to the touch. The characteristic odour of fresh meat should be present, and the meat-tissue should be dry, or at the most merely moist. The appearance of good meat is marbled, and its action on blue litmus-paper is acid—that is, it turns the blue paper to a red colour. Bad meat, on the other hand, is usually extremely moist, or even wet; it has a sodden feel, and the presence of dark spots in the fat is a suspicious sign. The marrow of the bones, instead of being light red in colour, as in fresh meat, is brown-tinted, and often shows black spots. Tested by litmus-paper, bad meat is either neutral or alkaline, and turns red litmus-paper to blue, or does not alter either red or blue test papers. The odour of bad meat is highly distinctive; and its colour, as a rule, is suspiciously dark.

Regarding those animal-diseases which are believed to unfit the flesh for human use, considerable diversity of opinion exists. For example, the flesh of animals suffering from *pleuro-pneumonia* is regarded, almost universally, as unfit for consumption; although opinions exist which regard such flesh as harmless. Here, as in the case of milk, already alluded to, it is probable diversity of opinion arises from the different conditions under which the results of eating such flesh have been studied. In some cases, it is true, no evil results have accrued from this practice; Loiset showing that

during nineteen years, at least eighteen thousand oxen suffering from pleuro-pneumonia were killed and used in Lyons, as food, without any known evil results. But it should be remembered that the disease has its advanced as well as its initial stages; and in any case the opinions expressed with regard to the harmless character of the flesh, can only apply to cases in which the animals have been killed in an early phase of the disorder. The disease known as 'braxy' in sheep presents a similar conflict of opinions. Over fifty per cent. of young sheep in Scotland are stated by Mr Cowan in his Essay (1863) to perish from this disease. The disorder is a fever, attended by very characteristic symptoms; but 'braxy mutton' is eaten nevertheless by Scottish shepherds with impunity—although an important precaution is observed in this case by steeping the mutton in brine for six or eight weeks, and then drying it. The chief danger which appears to arise in man from the use of diseased meat is the development of blood-disorders and of blood-poisoning. 'Carbuncular disease' has increased in Scotland since 1842, when pleuro-pneumonia first appeared; and this affection has apparently increased since lung-diseases in animals have become common. On the whole, then, it may be urged that even with opinions of weight which allege the harmless character, in certain cases, of the flesh of diseased animals, there are risks involved which make the rule, that meat under such circumstances should be rejected, a highly safe and commendable practice both for public and trade attention.

In the case of the *parasites* which may affect meat under certain circumstances, there is fortunately no diversity of opinion to be encountered. The question of 'braxy mutton' may be debatable; in that of meat infested with parasites, no argument is permissible. All parasitic animals are liable to induce disease of more or less serious character in man; hence, if meat can be proved to be so infested, it should be summarily rejected.

The most common parasites which man is liable to acquire from flesh of various kinds are certainly *tapeworms*, which have been frequently described, and the dangers from which are well known. More serious in its nature is the *Trichina spiralis*, a minute worm, found chiefly in the muscles of the pig. This worm, if eaten by man with pork, develops with great rapidity within the human digestive system, and produces enormous numbers of young, which, boring their way through the tissues to the

muscles of the patient, cause serious and often fatal illness. Once in the muscles, no further change ensues to the worms, which simply degenerate into mere specks of lime. It is this *trichina* which produces the disease known as *trichinosis*. Fatal epidemics of this disease are not uncommon on the continent, especially where the unsanitary practice of eating uncooked or dried sausages is greatly in vogue.

Regarding the prevention of the diseases caused by parasites, one stringent rule should be invariably kept in mind—namely, that all flesh-meat should be *thoroughly cooked* before it is consumed. The practice of eating underdone meat and smoked provisions is attended with great danger. A degree of heat sufficient to cook meat thoroughly, may, as a rule, be trusted to destroy parasitic life which the flesh may contain—although, of course, no one would sanction the employment as food of any meat known to be parasitically infested. To this necessary precaution may be added the advice, that drinking-water should never be taken from ponds, lakes, canals, or rivers in which vegetable matter grows freely, as such water is liable to contain parasitic germs; and all vegetables used for food, and especially those used raw—as in the case of salads—should be thoroughly washed before use. Our dogs being liable to harbour certain forms of internal parasites highly injurious to man, should also have their health and feeding inspected and supervised. And it may be lastly mentioned, by way of encouragement in sanitary reform, and in the care and selection of our flesh-foods, that as far back as the reign of Henry III. the desirability of securing meat free from parasites was clearly known. In the reign of that monarch, butchers who were convicted of selling ‘measly pork’ were sentenced to exposure in the pillory as a punishment for their misdeeds.

BY MEAD AND STREAM.

CHAPTER XXXV.—THE MAID WAS IN THE GARDEN.

MADGE was glad that it was in her power to comfort Philip, most glad, because, in spite of the relief which he found in her presence, a vague fear was beginning to creep into her mind that somehow this power was slowly weakening. Was it his fault or hers? Was it the knowledge that the confidence which they had desired to keep perfect between them was no longer perfect? Was it the knowledge that she had accepted a secret which could not be shared with him that, disturbing her mind, suggested changes in him which had no existence? Maybe, maybe, and yet ... relieved as he had been for a little while, there was no mistake, there was no mistake about the weary look in his eyes when he was going away, or about his nervously lingering manner of saying 'Good-night,' as if he were afraid to leave her, lest the bogeys which had arisen in his path should seize upon him the moment he should be alone.

She had many bitter reflections that night before she went to sleep: first, about the position in which she was placed against her will; and next about the customs which allowed a woman so few opportunities to give practical assistance to the man she loved. If he had been only a labourer and she a washerwoman, then she could have been of some real value to him. As it was, she must stay at home, await his coming when the struggle was over, give him sympathy when he was in difficulty, and nurse him when he was sick. That was all. She wanted to be by his side in the heat of the struggle, helping him with hands and head as well as heart. She wished that his enterprise had assumed some other form than its present one, so that she might have had a full share in the actual work of it. To her it was absurd that, because she wore petticoats and happened to be above the necessity to earn a living, she should be excluded from his office, or go to it under the penalty of bringing ridicule upon him. She knew how many times in those weary chambers, and in that weary office during this period of worry and disappointment, he must long for her to cheer and steady him as only she could do.

As for Wrentham, she had not much faith in him, although, having no specific charge to make against him, and aware of Philip's confidence in him, she remained silent. She could only have said: 'I do not like him;' and Philip would have laughed at her, or chid her for being ungracious to his friend. She had not forgiven Wrentham for the accident with the horse; and she was not yet satisfied about it, for she could not forget what Uncle Dick had said in his passion.

'If I wanted to kill anybody, do you know what I'd do?—that is, supposing I could go about it in cold blood. Well, I'd keep a mettlesome mare in the stable for three or four days, feed her high, and then ask the man I wanted to hurt to take a ride on her. Five hundred to one but he'd come back in a worse plight than Philip did. And that's what I'd have said the man was trying on, if they hadn't been such close friends.'

Uncle Dick did not repeat this angry exclamation; but Madge could not forget it, and the remembrance of it made her this night the more discontented that she could not be always with Philip during the ordeal through which he was passing.

However, there was one way in which she might render him practical assistance; that was, by setting Caleb Kersey's mind at ease, and so enabling him to serve his master with a light heart, which is always a brave one. She had delayed speaking to Sam Culver until she could tell him that Caleb was not only working steadily but was successful, and could offer Pansy a comfortable home. She would not wait any longer: she would speak to them both in the morning. That thought helped her to sleep. For the time, the more serious business which she had to do with Mr Hadleigh held only a distant place in her mind.

Caleb had not been making progress in his wooing, and when he became aware of that fact, he grew discontented with the nature of things in general and especially with himself. The discontent with the condition of his fellow-labourers which had earned for him an ill repute amongst the farmers, had some grains of reason in it. There was no doubt that the majority of the labourers had large families and scant fare; that their cottages were in many

instances examples of the deplorable state of ruin into which roof and walls may fall and still be reckoned fit for human habitation; whilst in harvest-time, when there was an influx of labouring men, women, and children from the large towns and from Ireland, the lodging arrangements were disreputable. But in the present case, he could discover no reason to justify his discontent, and that made him feel bad.

He had never been a regular churchgoer, and for some time he had ceased going altogether; but lately he had become so punctual in his attendance, that the beadle-sexton, the clerk, with old Jerry and young Jerry Mogridge, had held more than one consultation on the subject in the taproom of the *Cherry Tree*. They shook their heads very wisely, and thought that there must be something wrong about this sudden conversion. But the vicar, who had as quick an eye for every face in his congregation as the thorough shepherd has for every sheep in his flock, was pleased, and concluded that there was some good spirit at work in the Agitator's mind. He would not speak to him yet. He knew how easily a hesitating sheep may be frightened away by over-zeal on the part of the shepherd. He would wait until the man felt quite at his ease.

So, in a distant corner of the church, Caleb sat Sunday after Sunday, his eyes fixed on the back of Pansy's hat, and brightening when any of her movements enabled him to catch a glimpse of her face. At first he merely dawdled along the road in the wake of Pansy and her father on their way home, until they entered the gates of Ringsford. There it was Sam's custom to halt and gossip with the gatekeeper; whilst Pansy hastened home by a bypath through the trees, in order to have dinner ready for her father. Then Caleb, by hurrying to the home-field and crossing it, would catch another glimpse of her before she entered the cottage.

He was ashamed of dogging their steps in this fashion, and could not help himself. Several times he made up his mind to speak to the gardener, and find some excuse for walking along with them; but he could not yet muster courage to grasp so much joy, although it was well within his reach. One bright day, however, he was as usual standing in the porch to see Pansy as she went out, and receive from her as usual a bashful glance and timid smile, which made the food he lived on for the week, when he was almost startled by her father speaking to him:

‘Come up the road a bit wi’ us, Kersey, if you have naething better ado.’

Caleb muttered that he was ready, and muttered still more awkwardly to Pansy that he hoped he saw her quite well.

‘Quite well, thank you,’ was the demure reply; and there was no further conversation.

She took her place on one side of her father, Caleb walked on the other. But she was there quite close to him, and—although decidedly ill at ease—he began to feel a degree of content which he had not known for many days.

The gardener had been amongst those who had observed Caleb’s conversion in the matter of church attendance, and being already sensible of the young man’s intelligent appreciation of his flowers, he was willing to credit him with having turned over a new leaf, and had charitably set aside his doubts of him.

‘Man, Kersey,’ said Sam, as soon as they were free from the crowd, ‘I have got one of the bonniest geraaniums that ever mortal set een on, and I want you to see it for yoursel’. I wouldna have asked you to come on the Sabbath, if it hadna been that I can never get sight of you on a week-day noo.’

‘I don’t suppose there can be any harm in looking at the flower,’ said Caleb, restraining the much more decided opinion he would have expressed on the subject if Pansy had not been there, or if he had been able to guess what she might have thought of it. One strong principle of his creed was that the more beautiful things men look at, the more refined their natures will become, and that for this purpose Sunday was the most appropriate day.

‘That’s just my opinion,’ was the satisfied comment of the gardener; ‘and I wonder you that’s fond o’ flowers, dinna take to studying them in earnest. Do you know anything at all about botany?’

‘Nothing,’ was the honest and regretful reply, for it was not easy to confess absolute ignorance in her presence.

‘Then you’ll just have to come whiles to see me, and I’ll learn you something about it. You will have to come especially in the spring-time; and it’s wonderful how soon you’ll find a real pleasure in it—especially in the geraaniums.’

In this way Caleb became a prospective pupil of the gardener, and after this he walked home with the father and daughter every Sunday. And Pansy became more and more shy in his presence, and blushed more deeply at his coming; whilst his heart swelled and throbbed, and the words he wanted to speak played tantalisingly about his tongue, but found no voice. By-and-by there was a curious change in Pansy. Her shyness and her blushes disappeared: she spoke to him in much the same manner as she did to Jacob Cone or Jerry Mogridge or any of the other men about the place. At first he was disposed to be pleased with the change, for it seemed to make him more at home when he visited the cottage. Presently he began to fancy that she tried to keep out of his way, and he did not understand it. Then one day she had a basket of flowers to take up to the house for the young ladies, and Caleb accompanied her. As they neared the house, he surrendered the basket to her, and he had only done so when they met Coutts.

‘Ah, early birds!’ he said, with his cynical smile; ‘good-morning.—Will you give me a flower for my button-hole, Pansy?—Thank you. That is a very pretty one—it will make me think of you all day.’

He passed on, and Pansy was blushing as she used to do when Caleb spoke to her.

Caleb drew a long breath, and with it inhaled the poison which distorted all his thoughts. He spoke no word; but the gloom which fell upon him spoiled him for work, and checked his visits to the cottage until he heard that warning cry from Philip:

‘Trust her, man; trust her. That is the way to be worthy of a worthy woman.’

The words seemed to rouse him from a wretched nightmare and to clear his eyes and head. The words kept ringing in his ears, and when he peered through the black span which lay between this day and the one on which Pansy gave Coutts Hadleigh the flower, he felt that the darkness was due to films on his own eyes, not to change in the atmosphere.

He straightened his shoulders and raised his head: he was able to look his future in the face again.

‘I will trust her,’ he said to himself bravely. When he went to Gray’s Inn in obedience to his master’s instructions, he had only to say: ‘Thank you, sir; you have done me a deal of good, and I’ll do what you tell me.’

‘Spoken like the sensible fellow I always believed you to be,’ rejoined Philip, much relieved. He would have rejoiced, but he was at the time too much distracted by his own affairs to be able to feel elated by anything. ‘There will be no more sulks, then, no more losing heart and seeing mountains in molehills?’

‘I hope not.’

‘That’s right; and ... look here, Caleb. I have a notion, from something you said, that I know the man you have been worrying yourself about. Take my word for it, if my guess is right, he is much too cautious a fellow—to put it on no higher ground—and too careful of himself, to be a poacher. He likes a joke, though; and if I were you, I would not let him see that he was making me uneasy. You understand—he might for the fun of the thing get up some hoax.’

Caleb thought he understood, and at anyrate the main point was quite clear to him—he was to trust her. And he kept faith with himself in that respect. Whenever she seemed cold to him, he blamed himself for bothering her at the wrong time. She had other things to take up her attention—all the work of the cottage, many odd jobs to do for her father, besides the hens to look after and their eggs to gather for the breakfast-table of the Manor. When she seemed to be trying to keep out of his way, he set it down to the fact that she had something particular to do. He found excuses for every change, real or imaginary, that had come over her manner of treating him. Come what might of it, he would trust her.

Then there was a bright forenoon on which Philip sent him out to Ringsford to fetch a small box, and he had an hour to spare before he had to start for his return train. So he went over to the cottage. The sun was gleaming whitely on the little green in front, and the grass was sparkling with frozen dewdrops. There was Pansy—eyes in their brightness rivalling the flashing dewdrops, cheeks aglow with healthful exercise, and sleeves tucked up above the elbows—hanging out the clothes she had just taken from the tub.

Caleb halted at the corner of the green. He had never in this world seen anything so graceful as that lithe figure moving actively about in the clear sunlight casting the clothes over the lines, now reaching up on tiptoe to place a peg in some high place, and again whipping up her basket and marching farther along with it.

She had covered one long line and taken a clothes-pole to raise it. That was a feat of strength, and Caleb sprang to her side.

‘Let me do that for you, Pansy.’

‘Gracious!’ was the startled exclamation; and at the same moment he planted the pole upright, the clothes thus forming a screen between them and the vine-house where Sam Culver was at work.

‘You didn’t expect to see me here at this time of day,’ he said, laughing, but already beginning to feel awkward, and looking everywhere except where he most desired to look—in her face. ‘I had to come down for this box; and as there was time enough, I thought I’d come round this way.’

She laughed a little, too, at her scare, and then began to hang out more clothes on another line as hastily as if she had not a minute to spare. He looked on, his eyes glancing away whenever she turned towards him. She also began to feel a little awkward, and somehow she did not fasten the pegs on the line with such deft firmness as she had done before he made his presence known.

‘Father is in the vine-house,’ she said by-and-by, compelled to seek relief by saying something.

‘I wish you would let me do something for you,’ was his inconsequent reply.

‘Something for me!’

‘Yes, carry the basket—anything.’

‘The basket is empty, and I have to go back to the washhouse.’

‘I will go with you.’

‘But there is nothing to do except wring out the clothes.’

‘Let me help you with that.’

‘Pretty work it would be for you!’ This with a nervous little laugh, which she evidently intended to convey an impression of good-natured ridicule.

‘It doesn’t matter what it is, so being it is for you.’

She stooped quickly, seizing one handle of the basket; he took the other, and they lifted it between them. He looked straight in her face now, and he fancied that the colour faded from her cheeks.

‘Father is in the vine-house,’ she repeated, looking in another direction.

‘I want to tell you something, Pansy.’ He was a little husky, and unconsciously moved the basket to and fro.

She knew what he wanted to tell her, and she did not want to hear—at least not then.

‘I can’t stay—I must run in now.’ She tried to take the basket from him.

‘Don’t go yet. I made up my mind to tell you when I was standing over there looking at you. I was meaning to do it many a time afore, but just when I was ready, you always got out of my way, and I couldn’t say it when you came back.’

‘I wish you’d let me go. I don’t want to hear anything—I’m in a hurry. Won’t father do?’

She was nervous; there were signs even of distress in her manner, and she could not look at him.

‘Ay, your father will do,’ he answered earnestly, ‘if you say that I may tell him we have agreed about it.’

‘About what?—No, no, no; you must not tell him that. We are not agreed. We never will agree about *that*.’

She was frightened, dropped the basket, and would have run away, but he had caught her hand. He was pale, and although his heart was hammering at his chest, he was outwardly calm.

‘Don’t say never, Pansy,’ he pleaded in a low voice; and she was touched by the gentleness of it, which contrasted so strangely with the manner of the loud-voiced orator when speaking to a crowd on the village green. ‘I’ve scared you by coming too sudden upon you. But you’ll think about it, and you’ll give me the right word some other time.’

‘There is no need to think about it—I cannot think about it,’ she answered with tears of mingled vexation and regret in her eyes.

‘But you’ll come to think about it after a bit, and I’ll wait—I’ll wait until you come to it.’

‘I never will—I never can.’

‘You’re vexed with me for being so rough in my way of asking you. I couldn’t help that, Pansy: but I’ll be patient, and I’ll wait till you come round to it or ... until you say that you can’t do it because your head is too full of somebody else.’

Pale and earnest, his lips trembled as these last words passed them. She uttered a half-stifled ‘Oh!’ and ran into the cottage. He stood in the bright sunlight looking after her, and the gloom fell upon his face again. There was something in that cry which seemed to tell him that her head was already too full of somebody else for him to find the place he yearned to hold in her thoughts. He knew the somebody.

(To be continued.)

THE CHARR OF WINDERMERE.

THE confined localisation of this delicate fish renders its natural history somewhat difficult to ascertain. As little, or even less, is known of its proceedings during a great portion of the year as of the salmon itself during its sojourn in the sea. There are several varieties of the charr in the Lake district of Cumberland, Westmoreland, and Lancashire; but undoubtedly they are merely the same fish changed by circumstances and general surroundings; just as the common trout varies in appearance, size, and condition according to the nature of the water in which it is found and the food obtained there. Charr are found in many of the Scotch and Irish lochs; whilst in the English Lake district they are more or less plentiful in Windermere, Coniston, Buttermere, Hawes-water, Ennerdale, Crummock-water, Goats-water, and one or two other tarns or meres. In the first-named lake it is by far the most numerous; and Coniston holds a good supply, though Sir Humphry Davy, writing thirty years ago, says: 'The charr is now scarce in Coniston, and quite extinct in Ullswater.' Now it is occasionally found in the latter; whilst in the former it is plentiful, and, with a comparative discontinuance of the working of the lead mines, the wash from which polluted the water, is increasing. Large quantities of very fine fish were taken during last year. It is similar pollution which has destroyed the fish in Ullswater. For this beautiful lake, let us hope for a return of the olden times, when charr and trout and skellies 'peopled' its waters, over which the kite and golden eagle often flew, and down whose slopes the red-deer from Martindale fells may even now find its way to quaff a morning's draught. As regards edible qualities, the Windermere and Coniston charr are the best; those of Hawes-water and Goats-water being smaller and of inferior quality.

Local history tells us that the love of a dainty dish induced the monks of Furness to stock Windermere with charr, which were obtained from some lake in the neighbourhood of the Alps; hence the fish is still known as *Salmo alpinus*; but the correct nomenclature is *Salmo umbla*. The same history or tradition tells us that this fish was placed there only about two centuries ago. Against this, a manuscript has recently been discovered,

bearing date 1535, to the effect that a certain Jacques Tallour was permitted 'to catch and tol the fayre fish charr in Wynandermer, and also his son Gerald.' There is no reason to doubt that the charr is as likely to be indigenous to some of our lakes as our ordinary trout. During a considerable portion of the year, the charr frequent the deepest parts of the lake, feeding upon and finding nourishment in the minute crustaceans and larvæ found in such places. In this respect the nature of this fish is actually the reverse of that of the trout, which delights in the shallows, and feeds on the flies and moths hatched on the gravel-beds and elsewhere. Nature would doubtless 'people' Windermere, Coniston, and other lakes with that fish which could best live in its deepest parts, and this fish is the charr. Probably, specimens were removed from here to smaller sheets of water, in some of which, however, it fails to thrive, though breeding and increasing in numbers. There is a vast difference in appearance between the charr of Windermere and the charr of Hawes-water: the latter thin and flabby; the former elegantly shaped, and more graceful in outline than the trout, not so fat and podgy as many of our spotted beauties are; a general and a uniform shade of pinkness appears, as it were, to shine through the skin; in some specimens, as it approaches the belly, this hue becomes a deep red; hence the 'red-bellied charr.' It has, of course, other distinctive differences, as in the shape of gill covers, number of fin rays, &c., which have often been described.

Unfortunately, our charr is mostly a bottom or mid-water feeder, and cannot take high rank as a sporting fish; but on the table it excels. In size it varies from a pound in weight downwards, though larger specimens have often been caught. The usual size is about three fish to the pound of sixteen ounces; though in Hawes-water and Goats-water, about eight to the pound is considered the usual run. In both these tarns the charr rises pretty freely at the fly, indicating an insufficiency of food below the surface; and it is this bottom-food which gives to them the excellent condition and flavour they attain in the deeper and larger lakes. The same may be said of the gillaroos, found in some of the Irish lochs.

It is surprising that more attention has not been given to the artificial rearing of charr. Some years ago, the Windermere Angling Association hatched and turned into that lake some thousands of the young fish; but the earliest note we have of their artificial rearing was by Dr Davy, then living at Lesketh How, Ambleside. This took place about thirty years ago, and was done in

the most rough-and-ready fashion. Still the infant fish were produced from the milt-impregnated ova; and a few days after hatching, and with the 'sac' still in attachment, the delicate 'infants' were transferred to Easedale tarn. Too young to defend themselves, the fry no doubt perished. Yarrell says that in the autumn of 1839, several charr, of some half-pound weight each, were placed in Lily Mere, not far from Sedbergh. Twelve months later, two of these fish, when retaken, were said to have been two pound-weight each! They were served at the Queen-dowager's table at Kirkby-Lonsdale. These reputed large charr were no doubt trout, for which the mere in question was famous. A few years since, charr were placed in Potter Fell tarn, which is connected with the river Kent (Westmoreland) by a small runner. One of these charr was caught with fly in the river itself, some miles from the tarn. It had increased in size from about four to some seven ounces in the space of twelve months. It was kept alive, and in due course returned to the Potter Fell. This is evidence that charr may live in a stream, and in the absence of suitable bottom-food, adopt the habits of the trout, and rise to the fly. On this account, they are worth cultivation; and their delicacy and fine flavour make them more valuable than the best trout—a fact which should be an inducement to their propagation. Potted charr is considered amongst the greatest fish-dainties that can be set before the gourmet.

The charr is usually taken in nets, though often caught with artificial baits, trolled at varying depths, after the style of the paternoster used in perch-fishing. Commencing at the beginning of March, the fishermen know the water the charr frequent, and soon find at what depth they lie in shoals or schools. As the season becomes warmer, the charr approach nearer the surface; and in genial weather, towards the end of May or beginning of June, are at times seen basking near the surface of the lake; not feeding, but 'bobbing' their noses out of the water, causing rises or bubbles, which in calm weather are easily discerned by the fishermen. If possible, the shoal is surrounded by a net or nets, and a rare capture ensues. Upwards of one hundred and eighty pound-weight of charr has thus been taken at one haul; and when one considers they are worth wholesale from sixteen to eighteen pence per pound, the employment cannot fail to be a lucrative one. We cannot, however, commend the practice of netting, which is not sport, but wholesale destruction.

SILAS MONK.

A TALE OF LONDON OLD CITY.

IN FOUR CHAPTERS.—CONCLUSION.

THE streets in the old city are dark and deserted as the detective and Walter Tiltcroft hasten through them towards Crutched Friars. The street-lamps cast limited spaces of light upon the fronts of lofty warehouses and counting-houses, leaving limitless spaces of shadow about and above. The windows of these mansions have the blankness of blind eyes; the great, black, massive office-doors are firmly closed; and the greater doors of the warehouses are fastened with huge padlocks and chains, like prisons, or places with dead secrets made safe in the custody of night. Not a word is spoken. The two men, earnestly bent on their search, walk along with the echoes of their footsteps sounding loudly in their ears; while the tap on the pavement of Fenwick's stick falls with a musical ring, as though it were gifted with the power, like a magic wand, of chasing the echoes away. When they presently stop at the entrance to the counting-house of Armytage and Company, the detective produces a latchkey, opens the door, and leads the way into the house. As soon as Walter has entered and the door is closed behind him, Fenwick draws forth a dark-lantern, which he flashes unceremoniously in the young clerk's face. 'I call this light,' says Fenwick, 'my eye.'

Walter stares at it, and blinks.

'It has peered into and pierced through many a dark deed.—Catch hold!'

Walter, with trembling expectation, takes the lantern.

'Throw the light upon the keyhole!' cries Fenwick. 'I will open the door.' He rattles, as he speaks, a bunch of keys.

'Which keyhole first?' Walter asks.

'The strong-room.'

Walter shows the way. They pass through the clerks' office and reach the iron-bound door of the strong-room. The keyhole is rusty with age; and when Fenwick stoops and applies the key, there is a grating sound inside the lock like the grinding of teeth. As soon as the door is thrown open, Walter, with quick-beating heart, flings the light forward into the room; that strange fancy coming over him that his eyes will encounter the ghostly form of the old miser, as he had imagined him that afternoon, wrapped in the white shroud, dancing round his heap of gold. But finding nothing except dark walls, he boldly steps in. The high stool beside the old desk, where he has so often seen Silas Monk sitting and poring over large ledgers, is vacant, and the ledgers are lying about on the desk, closed.

'Now,' says Fenwick, 'give me the lantern.'

Walter complies, and the detective flashes the light about from ceiling to floor. Suddenly the two men are startled by a stifled cry. Fenwick casts his lantern angrily upon Walter's face, as though he suspects him of having uttered it. The clerk's eyes are terror-stricken, and his face deadly pale.

'What's that?' asks the detective.

Walter clutches at Fenwick's wrist. 'It is the cry which I heard this afternoon.'

'What do you mean?'

The light of the lantern is still on Walter's face as he answers: 'I was seated at my desk. The cry came from this room; but I thought it was a fancy. At that moment Mr Armytage sent for me, and I was afraid, if I mentioned it, that the clerks would laugh at me.'

'Why?' asks Fenwick, with surprise. 'Do you believe in ghosts?'

'N—no,' says Walter with some hesitation. 'But that cry did seem rather ghostly too.'

'Nonsense! It is Silas Monk.'

'But it sounded,' continued Walter, 'as though it were in this room.'

'That's true.'

'Then it must be his ghost; for there is no living being here except ourselves.'

Fenwick again flashes the light from ceiling to floor, as though to make sure of this. Then he says: 'Kneel down, my lad. Place your ear to the ground, and listen.'

Walter quickly obeys; and for some minutes a dead silence reigns in the strong-room. The beating of his heart is all that Tiltcroft hears; and all that he is otherwise conscious of is that Fenwick's 'eye' is watching the side of his face uppermost on the floor as he lies there listening. Their patience is presently rewarded. Their ears are filled with another cry, pitiable and more prolonged.

Walter springs to his feet. 'It is there!' he cries.

'Below?'

'Yes; directly beneath our feet.'

The detective begins to examine the flooring. Inch by inch the 'eye' wanders over the ground. An antique threadbare drugget is moved on one side; packets of papers, ledgers, and lumber are shifted from one corner to another. At last Fenwick lights upon a circular hole about the size of a crown-piece, scarcely an inch deep. 'Ah!' cries he, 'now we are on the track.' He takes from his pocket a penknife, scoops about, and turns up a ring attached to the floor. He puts his large muscular thumb into this ring, and gives a jerk. A patch three or four feet square in the boarding is detached. 'A trap-door!' cries Fenwick. 'Stand clear.'

So it proves—a trap-door, which the detective quickly raises, revealing pitch-darkness in the opening.

'Go below,' says Fenwick; 'I'll follow.'

Walter looks down, hesitating. But when the light is thrown that way, and he observes that there are steps leading into the obscurity, he takes the lead. The descent seems endless; for he moves slowly, as Fenwick, coming after him, throws the light upon him. Walter hears the hard breathing of the detective, and it sounds so strange in the stillness that he holds his own breath to listen. Suddenly the light from the lantern falls upon something which glitters on the ground on all sides.

'Gold!' cries Walter. His feet touch the ground. He stoops and picks up a handful of sovereigns. 'The place is a vault, and it is paved with gold.—'

What's that?' He points to something in one corner like a human form.

The detective steps forward and bends down, throwing the light upon a ghastly wrinkled face. The small eyes glitter like the gold, as though they had caught the reflection, and the long lean fingers are clutching sovereigns and raking them up. Fenwick touches the miser on the shoulder. 'What is all this?' asks he. 'Have you lost your senses?'

The old man utters a cry of distress which has in it a ring of madness.

'Speak to him, my lad,' says Fenwick. 'He will perhaps recognise your voice.'

Walter kneels and takes the old miser's hand. 'Mr Monk,' says he, 'do you know me? I am Walter Tiltcroft, your friend.'

Silas Monk looks up, bursts into a wild fit of laughter, and then falls back senseless.

The detective lifts the old man in his strong arms as though handling a child. 'Ascend the ladder!' cries he quickly to Walter, 'and show a light; not a moment must be lost in getting the old man home.'

Silas Monk was taken back to his tumble-down dwelling in the dismal row, and was tended with all possible care by his devoted grand-daughter. His recovery to a certain point was rapid. But the mental condition was curiously impaired. His brain had lost its force; no recollection of the past survived. His memory seemed to have fled into darkness, and to be resting there and sleeping—a darkness into which it was safer not to admit a single ray of light. This was the bitter irony displayed by nature when granting to this old miser a further extension to his lease of life. For time out of mind, Silas Monk had been governed by a master-passion—his only thought that of hoarding gold. The glitter, like sunlight, had pierced his cold heart, and had helped to keep it beating; and it would almost seem as though the warmth which this gold had driven into his veins still lingered there, and helped to sustain vitality, even when the memory which had given birth to all this agitation was dead.

It had been thought advisable by those who study the mysterious workings of the mind, that gold should be concealed from the sight of Silas Monk, and, if possible, even the sound of it, in order that his memory might rest dormant and his life be prolonged.

One evening the old man was seated in his armchair before the fire, with closed eyes. Rachel sat on a low stool at his feet, holding his hand. On the other side of the hearth was Walter Tiltcroft.

‘Walter,’ said the girl in a low voice, ‘you hardly know how happy I am, now that grandfather can give me all his love. He thinks no more about his’——She stopped, and looked up at her grandfather’s face, frightened that even the mention of gold should reach his ears.

‘Ah!’ cried Walter with a sigh, ‘how many are there, I wonder, in this old city whose minds would be less disturbed if that precious word was forbidden to be uttered in their presence? Does not your grandfather already look less pale and haggard than he did a few weeks ago?’

‘Indeed, he does,’ replied Rachel. ‘He remembers both of us when we are near him. He seems to need nothing now except our affection.’

Walter took the girl’s disengaged hand and said: ‘Rachel! Let me be near you and him. Why should we not be one, and watch over grandfather together?’

At the young man’s words, a look of rapture crossed the girl’s face. ‘Dear Walter,’ cried she, ‘that is all I wish for in this world!’ She spoke like a true and tender woman—from her heart. Seated there by that homely fireside, with the only two beings who were dear to her, she never thought, or cared to think, that all the gold which Walter Tiltcroft and the detective had found in the vault below the strong-room in Crutched Friars would one day belong to her—that, when her grandfather died, she would be a great heiress—worth, indeed, some thousands of pounds. All she thought of, with that look of rapture in her face, was that she had gained Walter Tiltcroft’s love.

Meanwhile, Joe Grimrood having been accused of the robbery in Crutched Friars, was tried, and convicted. Thereupon, he made a full confession. For

some days before committing the theft, he had watched Silas Monk from the scaffolding, after the rest of the workmen had gone. Through a chink in the old shutter he had observed every movement of the old miser. He had seen Silas Monk raise the trap-door which led into the vault; he had seen him descend with his lantern, and bring up bag after bag of gold, and pour it out on the desk before him. Watching in Crutched Friars, after having been shown to the door by Walter Tiltcroft, he had seen the young clerk leave the premises. Re-entering the house by means of a key which he had taken the precaution to forge, he had gone straight to the strong-room, where he had met with unexpected resistance. Silas Monk had displayed, according to Grimrood's statement, almost supernatural strength; defending his gold as a tigress defends her young ones, with a savage leap at the workman's throat. When utterly exhausted, Grimrood had carried Silas down into the vault and had closed the trap-door upon him. Then, having placed all the gold with which the desk was covered, into the bags, the burglar had decamped, making his way to the docks, and securing a berth on board an emigrant ship which was on the point of departure for the high seas.

Thus it happened that, but for the shrewdness and energy of the detective, Joe Grimrood would have started on a voyage to Australia with, as it appeared, nearly a thousand pounds in hard cash belonging to Silas; and the old miser himself would in all probability have been left to die in the vault under the strong-room in Crutched Friars, and 'the mystery of Silas Monk' would have remained a mystery to the present day.

All this occurred some years ago. Silas Monk is long dead; and Walter Tiltcroft, who married the old miser's grand-daughter, is now a merchant-prince. He purchased, soon after the death of Mr Armytage, a partnership in the great firm; and thus the gold which old Silas had hoarded up in Crutched Friars proved the means, to a great extent, of making Walter Tiltcroft's fortune.

SOMETHING ABOUT THE HONEY-BEE.

BY A BEEKEEPER.

To ascertain the kind of flower, plant, or shrub which the honey-bee mostly prefers, is worth care and consideration. Having been a keeper of bees for some years, I think it may be useful to make known the results of my experience and observations in Somersetshire, Hertfordshire, and Middlesex.

I will suppose that I have purchased a new stock and hive, bar-frame for preference, and caused it to be removed from the market-gardens around Middlesex to a country town in Hertfordshire. My bees on arrival examine their prospect, and what an estate-agent may call their 'outlook,' very minutely, going even over the walls and trees adjacent to their own hive, and taking trial-trips of flight into the air, straight up—very like the rising of a skylark from a field—and dropping again almost as suddenly. Having to some extent, after a day or two, mastered the topography of the district, they will, if on a warm day in February, commence upon the crocuses, and work only upon them—not, as some may suppose, dodge about irrespective of the kind of flower. Although the casual spectator may see bees upon every description of open flower upon one and the same day, yet they are winging their way from different hives. Our bees have commenced on the crocus. The day following this, they will try the common field dandelion; and the next, the white arabis of the garden culture. Then the black-thorn; later on, the currant and gooseberry blossoms, and the sweet 'may' of our hedgerows; and of trees—lime, palm, chestnut come next.

The hive should face the south, and the alighting-board occupy as free a space as possible. Water should be given, even during winter—inside, if frost is severe.

Some beekeepers suppose that colour attracts the bee; others, that they possess acutely the sense of smell; and much has been written on the subject. But our readers are to suppose that we are keeping bees between us, and that I am relating my own experiences, which point to this—the

preference of these intelligent insects for some plants over others. I have tried to educate my bees, by inducing them on certain days to gather from flowers presented to them in small bunches upon the alighting-board of their hive. In two instances I succeeded. One was with white clover, which I picked in a field a mile distant. This appeared to cheer the bees greatly, and drove away their listlessness and inactivity. After making an examination of my offering, they began work in earnest; and this stimulant had the desired effect of inducing an idle community to work well. The second experiment was much more demonstrative. Early in the morning, before the workers came forth, I placed by the alighting-board some bunches of alder-flower. I had shortly the satisfaction of seeing the outgoing bees return with little white trousers of pollen, and I watched their flight to an alder tree at a corner of the garden, not far from their hive. This was conclusive.

Now for some descriptions of preference shown by bees. I have grown garden-peas of various descriptions near my hives without inducing the bees to notice them. Yet they will greedily gather from French beans or scarlet runners the whole day, till long after sunset. In spring-time, the yellow gorse on uncultivated spots forms a very strong attraction for the honey-bees; yet they never touch the blossom of the laburnum, which to ordinary mortals smells much the same. The cultivated hyacinth they do not care about, although they gather from the wild sort in the woods and shady groves. Bees show great preference for the pollen of some sorts of lilies, yet are wholly indifferent to the lily of the valley. They gather from the field-daisy, yet are careless of the cultivated sort.

Stocks they prefer to pinks, and lavender to either; also the small flower of the borage delights them; yet wild foxglove possesses little charm. I have heard that bees like monkshood, and will gather from it, but I have never seen them do so. If they did, their honey would be poisonous. Bees are passionately fond of clover and certain vetches, and they will desert any garden flowers for such natural feeding. Wild thyme and heather, which improve the flavour of the honey, bees perfectly revel in. Garden primroses, they do not care much for; and auriculas, however gaudy in colour, form no kind of attraction. The polyanthus they have a languid liking for. I have seen the wild-bees attack the cowslip; but not the honey-bee of our hives. I saw a bee once upon a cultivated rose; it was only resting. I have likewise a

distinct remembrance of seeing many upon the wild-rose and dog-rose, wild clematis, honeysuckle, and blackberry blossom.

The situation of our hive cannot always be in such a flowery land; and the beekeeper will do well to study the different flora and trees in the immediate neighbourhood of his hive, and endeavour to supply any deficiencies of pollen-bearing plants, as well as to give a gentle hint to the inhabitants of his hives of any honey-bearing plant from which he especially wants them to gather. Of course, in wild heather districts, there is no need to resort to planting or sowing for the bees; they will in such places always take care of themselves. In Somersetshire, bees find honey from the many miles of apple-orchard stretching away to the mild county of Devon; and farmers well know that a good bee season, with a warm and early spring, means a plentiful show of fruit in the autumn for cider. In and around Middlesex, there are market and fruit gardens; and in Hertfordshire, grazing and clover lands, besides hedges lined with limes and hawthorn, and later on, honeysuckle.

It is always a good plan to send late swarms of the hive into heather-bearing counties; for the bees being young, and having every inducement to work for the approaching winter, will store better than hives which have been 'swarmed' and deprived of honey, the colonies of which are worn or fatigued with the long-continued gathering of a summer in more southern counties. It must likewise be remembered that bees cannot gather, or rather will not do so, late in the autumn, when the cold prevents them sealing over with wax the top of the cell.

And now, a last word as to the preference of our bees for certain flowers over others, which we would imagine, with our limited powers of the sense of smell and taste, would be preferred by these insects, and for which we have the greater amount of regard. I have seen, upon the approach of a bee to any flower, that it flies around the calyx almost always before alighting upon the flower itself. This is a cursory examination; and with its antennæ outstretched and quivering, it is evidently scenting the honey contained within. Should this prove a fruitful flower and of the flavour required, the bee settles on the centre of the stamen, and clutching it with its four front-legs, steadies itself with its longer outstretched two hindermost ones, and withdraws the nectar by its proboscis, the rings of the body assuming a vibratory motion the while. The bee's proboscis is a most important

instrument. It is composed of forty cartilaginous rings, each of which is fringed with minute hairs, having also a small tuft of hair at its extremity, where it is somewhat serrated. Its movement is like the trunk of an elephant, and is susceptible of extension and contraction, bending and twisting in all directions. Thus, by rolling it about, it searches out the calyx, pistil, and stamen of every flower, and deposits its nectar upon the tongue, whence it passes into the gullet at the base. The gullet or first stomach is the honey-bag. No digestion takes place here. In shape, it is like an oil-flask, and when full, contains about one grain. It is susceptible of contraction, and is so arranged as to enable the insect to disgorge its contents into the cells of the hive. A short passage leads to the ventricle or true stomach, which is somewhat larger. This receives the food from the honey-bag, for the nourishment of the bee and the secretion of wax. Dzierzon says that the honey which a bee can take into her stomach will enable her to subsist for a week under some circumstances, while under others she will die of hunger within twenty-four hours. This opinion of Dzierzon settles my conviction, that in the selection of the kind of food which will enable the bee to live longest, the true guide is to be found in the flowers for which it has the strongest preference.

BOOK GOSSIP.

THE Norman Conquest is one of the great outstanding and predominating facts in English history. It occasioned a sudden break in the life of the English people, and its influence is felt in their character and institutions even to the present day. A hundred and fifty years before that event, the long black ships of the Norse pirates entered the wide mouths of the Seine and the Loire, and their crews, the rudest of the rough barbarians of Denmark and Norway, sacked the towns and pillaged the churches of the country which was afterwards to be called by their name. They had no science, no arts, no culture. Their physical strength was their glory; and their weapons of war, their defence at home, served also as their passport into the lands of the stranger whom they plundered and slew. But they had a remarkable power of adaptation. However foreign to them the environment into which their hardy courage had brought them, they did not long remain untouched by it. Without losing their own native hardihood and fearlessness, they quickly absorbed into them the spirit of the peoples and institutions among which they had taken root; and before a century had passed over their heads in France, they had already become one of the great political forces of Europe. It was this people, brave, warlike, and with strong practical sagacity, who landed on the English shores in 1066, and shattered the Saxon arms on the slopes of Senlac. The battle at 'the hoar apple tree,' where Harold lay dead with the Norman arrow deep in his brain, marks the beginning of a new epoch in England.

The history of that great event, with its antecedents and consequents, has rarely been better told than it is by Mr Wm. Hunt, in the new volume of the 'Early Britain Series,' entitled *The Norman Conquest* (London: Society for Promoting Christian Knowledge). As compared with the work of Freeman, this is in bulk but a small book; yet it contains within it all that thousands of readers would desire to know of the history of the Conquest. The author is extremely well-informed on his subject, and his scholarly little book gives evidence not only of original research but of much original thought. The pictures he draws for us of the England that preceded the Conquest, and of the England that followed it, are sketched with a fullness and beauty of

detail which amply exhibit the capacity and preparedness of the author for the task which he undertook, and which he has executed so well. His extensive reading has enabled him to take advantage of the results obtained by all the best and more recent investigators in this section of European history; and the Northmen both before and after their descent on France, as well as the Saxon tribes and Danish hordes that scoured our coasts centuries before, are portrayed with a quick and living touch. Still more interesting is the story of the Normans after their taking possession of England; and the strange manner in which the Saxon head eventually conquered the Norman hand—the Saxon language and institutions arising in more than their original vitality and force out of the ashes, as it were, of a temporary death—is here narrated with admirable clearness and coherency. The book is one of the best of the very valuable series to which it belongs.



The same publishing house issues another learned little volume on *Anglo-Saxon Literature*, by Mr John Earle, Rawlinson Professor of Anglo-Saxon in the University of Oxford. It belongs to the series bearing upon 'The Dawn of European Literature,' and is rich with the results of the best modern scholarship on the early history and growth of our language. The time when Latin and Greek formed the chief essentials of learning is fast receding into the past, and these languages are having a place assigned them more consistent with the necessities of the modern world, which is not tolerant of the acquisition of a kind of knowledge that in great part is archaic and useless. Under the influence of this change, our own language is rising into an importance which it could never attain so long as it was regarded simply as a vulgar tongue, and the historical study of English is becoming one of the most popular as well as one of the most useful pursuits of our philologists. The great English Dictionary of the Philological Society is only one evidence of this; for individual scholars, during the last twenty years, have done not a little to lay bare to us the inner structure of our language, and the changes and modifications to which it has been subjected in the course of its long descent.

In the little work under review, Mr Earle states that Anglo-Saxon literature is the oldest of the vernacular literatures of modern Europe. The materials of this early literature are found chiefly in written books and documents; but they are found also in such subsidiary sources as inscriptions on

churches and church towers, sun-dials, crosses, and even on jewellery. One of the most remarkable in this last category is what is known as the Alfred Jewel. It was discovered in Newton Park, near Athelney, in 1693, and in 1718 had found its way to the Ashmolean Museum in Oxford, where it still is. It consists of an enamelled figure enshrined in a golden frame, with a golden back to it, and with a thick piece of rock-crystal in front, to serve as a glass to the picture. Around the sloping rim the following legend is wrought in the fabric: ÆLFRED MEC HEHT GEWYRCEAN ('Alfred me commanded to make'). 'The language of the legend,' says the author, 'agrees perfectly with the age of King Alfred, and it seems to be the unhesitating opinion of all those who have investigated the subject that it was a personal ornament of the great West-Saxon king.' Mr Earle traces the language from the Heathen Period—that is, from the time previous to the English conversion to Christianity, about 597 A.D.—down to the times that immediately succeeded upon the Norman Conquest, and gives examples of the language during these six centuries, with translations of the various passages adduced. All who have an interest in the study of the English tongue, and of the changes superinduced upon it by contact with other European vernaculars, will find Mr Earle's volume a ready and efficient guide.

THE MONTH: SCIENCE AND ARTS.

PROJECTS for cutting waterways across isthmuses follow one another with such amazing swiftness, and the project is in most cases so quickly followed by realisation, that it would appear that before many years have passed, all the available peninsulas of the world will have been operated upon and transformed into islands. Our French neighbours are at present discussing the feasibility of a gigantic undertaking of this nature, which, if carried out, will unite the Bay of Biscay with the Mediterranean. This projected canal, which is to be of such dimensions that the largest ships afloat can make use of it, is to have one entrance near Bordeaux, and the other at Narbonne. This short-cut across France will obviate the necessity of the tedious voyage round Spain and through the Straits of Gibraltar, and will undoubtedly be a boon to shipping, and especially to British vessels; but the scheme is at present only on paper. It remains to be seen whether the undertaking is possible; by which is meant, in these days of engineering marvels, whether it will pay.

Like most other canal projects, this one is by no means new; indeed, a canal already exists almost along the same line of route—namely, the Canal du Midi, which finds an outlet at Cette in the Gulf of Lions, and joins the river Garonne at its other extremity at Toulouse; the entire navigable distance from Bordeaux to Cette being three hundred and thirty-two miles. The existing canal only accommodates small vessels, and the entire journey is by no means a rapid one, for there are more than a hundred locks to be encountered, which gradually raise the boats to a level of nearly eight hundred feet above the sea. Whether the engineers of the new undertaking propose any novel means of battling with this difficulty of level, we do not know; but it will be readily seen that the undertaking has not the simplicity of a simple cutting, such as the Suez Canal presents. Another formidable obstacle to the work is the presence of certain rivers which flow right across the track. In the present case, these are crossed by aqueducts. But what would be the size and cost of aqueducts which would give passage to the

floating palaces which have taken the place of the small vessels of days gone by?

Coming nearer home, a project has been mooted for cutting a channel from the river Tyne to the Solway; and another across the low land which separates the Forth from the Clyde. It is true that in the latter case a narrow passage already exists; but what is required is—according to the opinion of a former President of the Liverpool Chamber of Commerce, who writes to the *Times* upon the subject—a channel which will allow the passage of our largest merchantmen and ships of war, so that in case of need the efficiency of our naval defences may be practically doubled. In case of war, the advantages of quick transport of our ships from one coast to the other is obvious, and may in a manner be compared to the undoubted advantages which we reap from being able to convey information quickly from place to place by telegraphic agency.

Some very interesting Roman relics have recently been unearthed in the bed of the river Rhone at Geneva, where some engineering works are in progress. The most interesting of these is a Roman altar furnished with an inscription to the effect that the writer, a certain soldier of the twenty-second legion, who had been shipwrecked in the waters hard by, had raised this altar to the god of the waves, Neptune, as a thank-offering for his escape from death. We have also to record a far more valuable find near Rome itself—at Subiaco, where several priceless statues supposed to have been sent by the Emperor Nero to that place for the decoration of his villa there, have been dug up. In Britain too, a Roman villa has just been laid bare at Woolstone, Berkshire, where, in addition to many tessellated pavements, several graves of the Anglo-Saxon period have been found. In London, our knowledge of the Roman city which lies beneath the busy metropolitan streets has been much enriched by numerous discoveries made during the recent excavations for the completion of the Underground Railway. There is little doubt that interest in things antiquarian is rapidly increasing on all sides. This is not only apparent from the attention which every fresh discovery receives, but is indicated in a most satisfactory manner by the circumstance that the University of Cambridge has given archæology a recognised position among the subjects for the classical tripos examination, and has just opened a Museum which will give an impetus to studies of the kind.

Although interest in matters archæological shows a healthy increase, we have to regret a decrease of interest in another important branch of knowledge. The Royal Geographical Society, which has just held its anniversary meeting, has had to deplore, by the mouth of its President, Lord Aberdare, that the Council have failed in their attempt to introduce the efficient study of geography into the curriculum of our great public schools, such as Eton and Harrow. Prizes have been offered; but there were few who cared to compete for them. This seems a very extraordinary state of things in a country which is always proudly pointing to its possessions as being so large that the sun must always shine upon some part or other of them. But the fault probably lies with the teachers more than with the pupils. The members of the Geographical Society evidently understand this, for they are now about to institute an inquiry into the systems adopted for geographical instruction in continental schools, from which, if all reports speak truly, we may well take a lesson.

Professor Monier Williams's recent lecture on India, delivered before the University of Oxford, was full of interesting particulars relating to the great progress in every way which that vast country had experienced under British rule. But perhaps the most interesting portion of his remarks was that relating to the new route to India which will probably be opened, and which it is expected will lead to great development of intercourse between our Eastern and Western possessions. This route will consist of a journey from London to Odessa; thence by steamer across the Black Sea to Batoum; then by Russian railway—a thirty-six hours' journey—to Baku on the Caspian; and a day's voyage across the Caspian to Michaelovsk. At this latter place is the terminus of the Central Asian Railway, which some months ago was complete for one hundred and forty-four miles, and which will eventually land the traveller at the gate of India—Herat. The journey from Calais to our Indian frontier will be possible in nine days, so long at least as we remain friends with Russia. Professor Williams considers that we shall be bound to extend our railway from its present limit at Quetta, through Candahar, so as to meet the Russians at Herat. He thinks that we can meet them there as friends rather than enemies; and all will agree in trusting that his words may come true.

During the past year, the progress made by the British Ordnance Survey has been greater than in any previous period, an area of more than two and a

half million acres having been mapped. It is expected that the survey of the entire kingdom will be complete by the year 1888, and that the publication of the maps will be finished two years later. A largely increased staff of surveyors and draughtsmen has been engaged to insure this acceleration in the work, and considerable time has been spent in instructing their assistants in their duties. The maps are reduced to the six-inch scale, and are reproduced by the zincographic process. All particulars of the work are contained in a recently published Blue-book.

The long-continued dispute as to the right of the telegraph department to erect posts and wires over our crowded city streets has at last been set at rest, and the Postmaster-general can, with certain restrictions, do much as he likes about the matter. The Telephone Companies, who are new-comers and have no statutory powers, have yet to fight the question. We must for many reasons deplore the circumstance that additions will still be made to the metallic spider-webs which cover so many of our fine metropolitan streets. It has been suggested that the lines could be made to follow the contour of the roads, and could be hidden under eaves and behind coping-stones so as no longer to offend the eye, or to present the risk of danger to life, which they now undoubtedly do. This innovation would doubtless mean a great deal of difficulty to telegraphic engineers, and would be naturally opposed by them, for there is a sweet simplicity about a suspended wire; but the gain to others would be great.

The International Health Exhibition, London, which follows so closely upon the Fisheries Exhibition, and occupies the same spacious site, bids fair to be a success, although it can hardly be expected to be quite so popular with the multitude as its predecessor. Still, there is much to attract the far larger part of the community who long for amusement rather than instruction, and as the financial success of the undertaking must be dependent upon such visitors, the caterers cannot be blamed if they have admitted within their walls many exhibits which, by the widest stretch of the imagination, can hardly be associated with the subject of health. For more thoughtful visitors, there are Conferences upon all manner of questions connected with Domestic Sanitation, questions of which the majority of people are at present profoundly ignorant. There will also be papers read upon the subjects of Meat-supply; Food-adulteration and Analysis; School-diet; School-life in Relation to Eyesight; Posture in

Schools; Epidemics in Schools; and numberless other matters of social interest. As these Conferences are under the care of different Societies and Associations, which exist only to increase our knowledge regarding the different subjects indicated, and which have in most cases been at work for many years, we may be sure that much good will accrue from these discussions. Following the procedure of the Fisheries Exhibition, a number of pamphlets will also be issued, dealing with the multifarious sections of the Exhibition.

Although, as we have more than once pointed out, the general adoption of the electric light for domestic purposes cannot be looked for in the near future, it can easily be installed for special occasions. An account has recently been published of a ball at a private house in London where the rooms were illuminated during the evening by one hundred and twenty incandescent lamps. These lamps were fed by secondary batteries, which arrived in two vans, and which were subsequently accommodated in an adjoining coach-house. The batteries had been previously charged at a place ten miles distant. This use for the light may possibly become common in cases where cost is not a matter of first consideration.

Another phenomenal diamond has fallen to the lot of a fortunate digger at the Kimberley mine, South Africa. Its weight is three hundred and two carats; but, unfortunately, it does not possess that purity of colour, or rather absence of colour, which is the first desideratum in a diamond. Its value is said to be about three thousand pounds; whereas the far smaller Porter-Rhodes gem, found in the same mine about three years ago, was valued by its owner at one hundred thousand pounds. But the popular notion is that the value of a thing is what it will fetch, and there are certainly very few persons in the world who would lock up such an enormous sum for the doubtful advantage of possessing such a thing.

A document, which should be widely known, was recently issued by the Board of Trade, in the form of a Report of the first year's experience of the Boiler Explosions' Act of 1882. This Act, we may remind our readers, provides that an inquiry should be held into the cause of every boiler explosion, with a view to their prevention if possible. The causes of the forty-five casualties of this description which were inquired into, and which resulted in the loss of thirty-five lives and injuries to as many more, were entirely preventable. One of the assistant-secretaries to the Board goes so

far as to say that ‘the terms “inevitable accident” and “accident” are entirely inapplicable to these explosions, and that the only accidental thing about many of them is that the explosions should have been so long deferred.’ The prevailing cause of the disasters is the unsafe condition of the boilers through age, corrosion, wasting, &c.; and a noticeable feature in many cases is the absence of any effort on the part of the steam-user to ascertain the condition of the boiler, and consequently of any attempt to repair, renew, or replace defective plates or fittings.

The authorities of Kew Observatory have undertaken a duty which will be hailed with satisfaction by all watchmakers and watchowners in the kingdom. They will undertake for a small fee to test the virtues of any watch left in their care, and with every watch so tested, will issue a statement of its going powers, under varied conditions of position, temperature, &c. They will also award to watches of superior excellence certificates of merit, which certificates will possess an equal value with documents of the same nature which have for years been granted by the Geneva and by the Yale College Observatories. The Swiss and Americans have long enjoyed these facilities for obtaining independent testimony as to the qualities of their watches, and it is only surprising that a movement has not been made before in this direction here at home; for English-made watches, in spite of foreign competition, are still much sought after.

A new method of dealing with road-sweepings and the contents of domestic dust-bins is now on its trial in New York, and seems to be very successful. The rubbish is carted, to the extent of forty loads a day, to a wonderful machine, which separates the paper, rag, iron, glass, coal, and cinder into different heaps. These are afterwards sold, with the exception of about four hundred pounds of coal and cinder, which are used for firing the engine attached to the machine. The remaining refuse—of no use to anybody, and too often, under existing systems, a possible source of disease—is reduced by fire to impalpable ash. It has been the custom in New York for many years to carry their rubbish out to sea and to discharge it outside the harbour. Pilots and others have long protested against this procedure, and affirm that the approaches to the harbour’s mouth are gradually being silted up by the accumulation of dirt thrown in. The experiment will be watched with interest by all those who acknowledge the importance of improved sanitation in our large towns and cities.

Moon's Patent Quicksilver-wave Gold Amalgamator is the imposing title of a clever machine which has been introduced to obviate the serious loss of gold which is inseparable from previously existing methods of treating the ore. From the discovery of gold in California in 1848 to the end of 1882, the value of the gold found there was nearly two hundred and thirty-seven million pounds sterling. It is said on competent authority that this vast amount is less than fifty per cent. of the gold known to be in the ore treated, more than half the precious metal escaping in particles so fine that the machines employed could not intercept them. In this new machine, the crushed ore, mingled with water, is thrown in small quantities into a moving wave of quicksilver, and not merely across a quicksilvered plate, as under the old system. The tiniest spangles of gold are by this means speedily absorbed by or amalgamated with the liquid metal, the two being afterwards separated by heat in the usual manner. In one mine where Mr Moon's machine is in use the increase of yield is estimated at forty pounds sterling per week, so it would seem that the cost of the appliance is soon repaid to its purchaser.

A very convenient combined seat and easel for the use of sketchers has lately been brought under our notice. It packs into a very small compass; it will hold a large picture; it fully justifies its name, 'The Rigid,' and actually weighs only four pounds. Its price is moderate, and it is to be had of Messrs Reeves, London.

Referring to a recent article in this *Journal* on 'Some Queer Dishes,' in which it was stated that the cuttle-fish is used for food in Japan and elsewhere in the Pacific, a Portuguese correspondent writes to us that in Portugal the cuttle-fish is used as an article of food. It is opened, and then dried; and may be seen hanging up for sale in the shops. The people, he remarks, consider it a delicacy; and it is, when properly cooked, very rich and nourishing.

OCCASIONAL NOTES.

NEW POSTAL ORDERS.

THE system of Postal Orders, instituted in 1881, has proved so successful, that it has been found desirable to make certain alterations and extensions therein, with a view to affording further facilities to the public for the ready transmission of small sums of money through the post. On the 2d of June, a new series of Postal Orders were issued, the former series being entirely withdrawn. The new Postal Orders are of fourteen different denominations, instead of ten, as formerly; and the amounts of the various denominations, together with the rates of poundage chargeable thereon, are as follows:

<i>s.</i>	<i>d.</i>	<i>d.</i>
1	0	0½
1	6	0½
2	0	1
2	6	1
3	0	1
3	6	1
4	0	1
4	6	1
5	0	1
7	6	1
10	0	1
10	6	1
15	0	1½
20	0	1½

There can be no doubt that these classes will prove extremely useful to the public generally, more especially as any amount of shillings and sixpences up to twenty shillings can be transmitted by means of only two of the above-named classes of orders. A novel feature, too, is introduced, whereby postage-stamps not exceeding fivepence in value are to be allowed to be affixed to the back of any one Postal Order to make up broken sums—a feature which, it needs not much of the spirit of prophecy to anticipate, will extensively be taken advantage of. By this useful concession, any sum up to a pound can now be sent through the post by means of Postal Orders, and in no case are more than two orders required to make up the exact desired amount. It will be noticed that the former twelve shillings and sixpence and seventeen shillings and sixpence orders are not included amongst the new denominations of Postal Orders; but their abolition will cause no inconvenience, as these two denominations were of all the orders of the old series probably the least used; and where such amounts are desired to be sent under the new series, they can be made up by using two orders, the poundage on which will be no more than is now charged for each of the denominations referred to—namely, twopence. In several cases, the poundage has been reduced, a benefit that will probably be the best appreciated of all. A ten shillings and ten shillings and sixpence order now only costs one penny; and the orders for fifteen and twenty shillings have been reduced to three-halfpence, instead of twopence, as heretofore. Compared with the former money-order rates, the Postal Order system is remarkably cheap, and on this score, will undoubtedly commend itself more than ever to popular favour; and it is extremely probable that for small sums the money-order system will in future be very little if at all used. Indeed, the Postal Order system, with its ready convenience and cheapness, seems likely to supersede all other methods of transmitting sums of a pound and under.

The Act under which these changes have taken place also authorises the issue of Postal Orders on board Her Majesty's ships, a boon that the seamen concerned will not be slow to appreciate. The system is also to be extended to many of the colonies as opportunity occurs. It is indeed now in operation in Malta and Gibraltar, where it has met with much popularity, owing to the fact, no doubt, that the same rates are charged on Postal Orders issued there as on Postal Orders issued in this country. If we compare these rates with those charged on foreign and colonial money orders, it can readily be

imagined that the system will be hailed with unmixed satisfaction by the colonies where it is shortly to be instituted.

NEW METALLIC COMPOUND.

Delta-metal, a new metal said to be not unlikely to rival steel under certain conditions, has, according to the *Hamburg Correspondent*, been lately submitted to the Polytechnic Association in Berlin. Delta-metal contains iron in addition to the ordinary constituents of brass. It takes on an excellent polish, and is much less liable to rust than either steel or iron. When wrought or rolled, it is harder than steel, but not when cast only. It can be forged and soldered like iron, but not welded. It melts at about one thousand seven hundred and fifty degrees Fahrenheit; and at from one thousand three hundred to one thousand five hundred degrees it is remarkably malleable, and in this condition can admirably well be pressed or stamped. For founding, it is also well suited. The price is somewhat higher than that of the better kinds of brass. It should be found specially serviceable for objects exposed to rust and requiring great hardness. At present—not to mention other cases—the small steamers for the exploration of Central Africa are being made of delta-metal.

HARBOUR OF REFUGE FOR EAST COAST OF SCOTLAND.

We have before alluded to the operations of the Committee appointed by the Government to take evidence as to the most suitable place for a harbour of refuge on the east coast of Scotland. The Report of the investigators has now been published, with their final recommendations. The towns and harbours of Wick, Fraserburgh, Peterhead, Aberdeen, Arbroath, and Montrose, are severally reported upon, and the advantages and disadvantages of each stated, with the result that the reporters unanimously recommend Peterhead as the site of the proposed harbour of refuge. That town is situated almost midway between the great natural harbours of the Firth of Forth and Cromarty Firth, and its bay is well adapted as a place of shelter. Its anchorage also is excellent, the bottom of the bay being of mud with a sandy surface, affording a good holding-ground. The harbour is to be constructed by Scottish convict labour.

SUBTERRANEAN FISH.

A fact of much interest to students of natural history is vouched for by Cavalier Moerath, a civil engineer, formerly of Rome, and now visiting this country. This gentleman has devoted much labour and attention to the improvement of water-supplies in Italy. In sinking for water with one of Norton's Abyssinian Tube Wells, he tapped a spring from which was pumped a tiny living fish. This fish had passed into the tube well through the ordinary perforations of about one-eighth of an inch. Examination proved it to have no eyes, clearly indicating that it belonged to an order intended to inhabit subterranean waters. The occurrence was certified to by two other gentlemen who were present when the fish was pumped up.

The site of the well is Fontano del Prato, near the old city of Cori, between Rome and Naples, and the depth is about seventy feet. The soil from which the fish came was fine sand. The strata passed through above this sand were volcanic loose earth, clay and water, other volcanic earth, rocks and sand, and clay. The temperature of the water was low—about forty degrees Fahrenheit. The water was fresh. The fish, we are informed, has been preserved in spirit, and is to be brought to England, when it will probably be exhibited at the Health Exhibition in London.

THE FASTEST PASSAGE ON RECORD.

This great feat has just been achieved by the Guion line steamer *Oregon*, which left New York on the 26th of April last, and arrived at Queenstown at 5.16 on Saturday morning the 3d of May, making the trip in six days sixteen hours and fifty-seven minutes, which is the fastest homeward trip yet recorded. This is the more remarkable from the fact that she had to traverse over a hundred miles at least out of her course to avoid the icebergs, those pests of the North Atlantic. Passengers who embarked at New York on Saturday the 26th April were landed at Liverpool on the evening of that day week. The *Oregon* is another of those naval masterpieces for which the industry and skill of Scotland are so justly celebrated, and is considered one of the finest steamers afloat. Her highest score of miles run in one day was four hundred and thirty-six.

A CANINE 'COLLECTOR.'

That dogs can be taught the performance of tricks or acts showing a remarkable amount of sagacity and intelligence, no one will pretend to doubt, for it is a fact patent to all. But that a dog could become a 'collector,' and a collector of money too, is at first sight somewhat startling. Yet such is the fact. A splendid and thoroughbred Scotch collie, known as 'Help,' has been actually trained as a collector of money for charitable contributions, or subscriptions, for the 'Orphan Fund of the Amalgamated Society of Railway Servants.' His tutor has been one of the guards of the night-boat train on the London, Brighton, and south-coast line. He is described as a dog not only of great beauty, but of gentle and winning ways, possessing marvellous intelligence and a generous disposition. In his capacity as collector he has travelled over the greater part of England, always returning home to the headquarters in the City Road, London, with the proceeds of his charitable efforts. Last year, he is reported to have crossed the Channel, having been taken over by the captain of the steamer *Brittany*, and introduced by him to Her Majesty's consul at Dieppe. In this port he is stated to have collected about six pounds ten shillings; and on returning home he seems to have made a rather profitable stay at Newhaven, where he collected nearly seven pounds. In February last it was reported in the newspapers that Help had been killed at a level crossing at Middlesborough, in Yorkshire, where he had been run over by an 'express' train. This, however, turns out to have been a mistake. A handsome Scotch collie was killed as stated, and as he resembled Help very much, the story got about that the canine 'collector' had lost his life on the line. But Help is at this moment actively following his charitable avocation, in which, we believe, he excites more interest than ever. And long may he continue to carry on his useful career of helping the fatherless and the afflicted. It would be interesting to know the plan or system employed for the dog's operations; in other words, how it is done. The animal must, of course, always be in charge of somebody, otherwise, when he had done a fair day's work in collecting money, there are numbers of unprincipled people who would speedily ease the collie of his subscriptions, if they did not take his life as well.

WILD-FLOWERS FROM ALLOWAY AND DOON.

BY ALEXANDER ANDERSON.

No book to-night; but let me sit
And watch the firelight change and flit,
And let me think of other lays
Than those that shake our modern days.
Outside, the tread of passing feet
Along the unsympathetic street
Is naught to me; I sit and hear
Far other music in my ear,
That, keeping perfect time and tune,
Whispers of Alloway and Doon.

The scent of withered flowers has brought
A fresher atmosphere of thought,
In which I make a realm, and see
A fairer world unfold to me;
For grew they not upon that spot
Of sacred soil that loses naught
Of sanctity by all the years
That come and pass like human fears?
They grew beneath the light of June,
And blossomed on the Banks of Doon;
The waving woods are rich with green,
And sweet the Doon flows on between;
The winds tread light upon the grass,
That shakes with joy to feel them pass;
The sky, in its expanse of blue,
Has but a single cloud or two;
The lark, in raptures clear and long,
Shakes out his little soul in song.

But far above his notes, I hear
Another song within my ear,
Rich, soft, and sweet, and deep by turns—
The quick, wild passion-throbs of Burns.

Ah! were it not that he has flung
A sunshine by the songs he sung
On fields and woods of 'Bonnie Doon,'
These simple flowers had been a boon
Less dear to me; but since they grew
On sacred spots which once he knew,
They breathe, though crushed and shorn of bloom,
To-night within this lonely room,
Such perfumes, that to me prolong
The passionate sweetness of his song.
The glory of an early death
Was his; and the immortal wreath
Was woven round brows that had not felt
The furrows that are roughly dealt
To age; nor had the heart grown cold
With haunting fears that, taking hold,
Cast shadows downward from their wing,
Until we doubt the songs we sing.
But his was lighter doom of pain,
To pass in youth, and to remain
For ever fair and fresh and young,
Encircled by the youth he sung.

And so to me these simple flowers
Have sent through all my dreaming hours
His songs again, which, when a boy,
Made day and night a double joy.
Nor did they sink and die away
When manhood came with sterner day,
But still, amid the jar and strife,
The rush and clang of railway life,
They rose up, and at all their words

I felt my spirit's inner chords
Thrill with their old sweet touch, as now,
Though middle manhood shades my brow;
For though I hear the tread of feet
Along the unsympathetic street,
And all the city's din to-night,
My heart warms with that old delight,
In which I sit and, dreaming, hear
Singing to all the inner ear,
Rich, clear, and soft, and sweet by turns,
The deep, wild passion-throbs of Burns.

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